4.1 Approximate KL

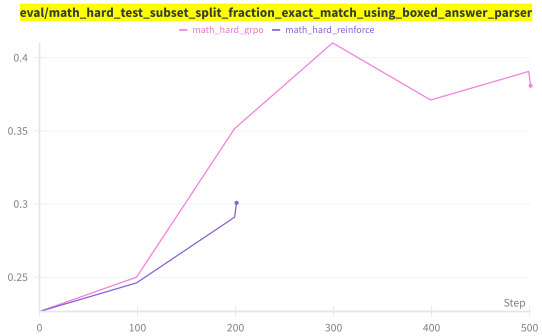
The estimator $\hat{k}(a) = e^{\Delta(a)} - \Delta(a) - 1$ is a valid sampled-token estimator for the KL divergence because its expected value over the sampled tokens is equal to the exact KL divergence. While the simpler estimator $-\Delta(a)$ is also unbiased, a single sample of it can be negative, which would incorrectly penalize the model in a way that behaves like a reward. The chosen estimator avoids this issue by remaining strictly non-negative, equaling zero only when the policy and reference probabilities perfectly agree. Computing the exact full-vocabulary KL at every token position would be vastly more expensive in both memory and compute, as it requires evaluating probabilities for every single token in the vocabulary at each step rather than just for the sampled tokens.

4.2 Implementation

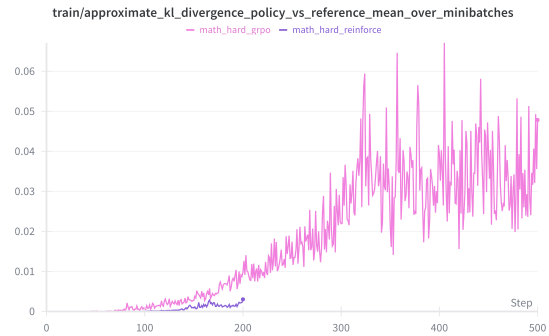
I implemented the code following the recommended order in Section 4.1 of the handout. One minor point of confusion during implementation was handling the edge case in `compute_group_advantages` when the `group_size` is exactly 1. I was initially unsure whether to return an advantage of 0 or fall back to the reward-to-go. I ended up choosing the former.

4.3 GR-REINFORCE vs. GRPO on `math_hard`

Over the first 200 iterations, the mean rollout reward is comparable between the two algorithms. However, GRPO shows faster improvement on the evaluation exact match metric, reaching approximately 0.35 at step 200 compared to 0.30 for GR-REINFORCE (Figure 4.1a). Additionally, the approximate KL divergence grows faster under GRPO, hitting 0.009 versus 0.003 for GR-REINFORCE (Figure 4.1b). This comparison is interesting because both commands use identical batch sizes, minibatch sizes, gradient accumulation steps, and learning rates. The observed differences highlight the impact of sample reuse: GRPO performs two clipped optimization passes over each rollout batch, allowing for more sample-efficient learning, whereas GR-REINFORCE updates strictly on-policy with a single pass.



(a) Eval Exact Match Ratio



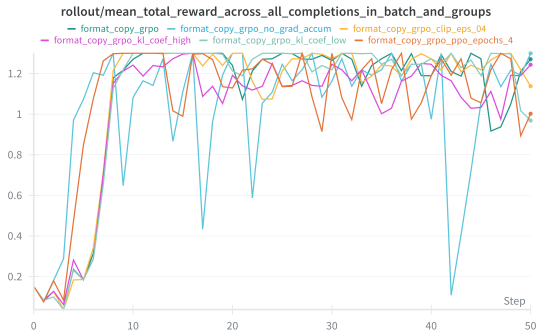
(b) Approx KL Divergence

4.4 GRPO Ablations on `format_copy`

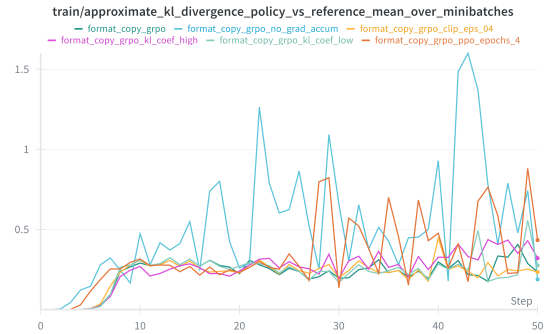
For the ablation study, I tested five different configurations:

1. Increased `ppo_epochs` from 2 to 4.
2. Decreased `kl_coef` from 0.05 to 0.005.
3. Increased `kl_coef` from 0.05 to 0.2.
4. Increased `clip_eps` from 0.2 to 0.4.
5. Decreased `grad_accum_steps` from 6 to 1.

All settings converged to the same evaluation exact match score. The hyperparameters that mattered most for training dynamics were `ppo_epochs` and `grad_accum_steps`. Specifically, the configurations with `ppo_epochs=4` and `grad_accum_steps=1` exhibited significantly higher variance in rollout reward (Figure 4.2a). While these two runs found high rewards earlier, they also produced the highest KL divergence from the reference policy (Figure 4.2b). This suggests that updating too frequently or reusing samples excessively makes the learning process more aggressive and prone to instability. The adjustments to `kl_coef` and `clip_eps` performed comparably to the default settings.



(a) Training Reward



(b) Approx KL Divergence

4.5 Qualitative Behavior

Below are two model generation examples from the `math_hard` task that illustrate the policy's progression during training. The first shows an early attempt where the model fails to format or reach a final answer, while the second shows a successful parsing of logic into the required format. Furthermore, I found that completions tended to be very long and verbose during early training steps, while solutions become more concise and neat during the later training stages.

Training Step 9: Early model generation failing to format or solve correctly

Question:

> Square $ABCD$ is inscribed in a circle. Square $EFGH$ has vertices E and F on $\overline{CD}$ and vertices G and H on the circle. If the area of square $ABCD$ is 1, then the area of square $EFGH$ can be expressed as $\frac{m}{n}$ where m and n are relatively prime positive integers and $m < n$. Find $10n + m$.

Model Completion:

> To solve the problem, we start by noting that the side length of square $ABCD$ is 1 since its area is 1. Therefore, the diagonal of square $ABCD$ is $\sqrt{2}$. Since square $ABCD$ is inscribed in a circle, the diagonal of the square is the diameter of the circle. Thus, the radius of the circle is $\frac{\sqrt{2}}{2}$.
>
> Next, we consider square $EFGH$ with vertices E and F on $\overline{CD}$ and vertices G and H on the circle. Let the side length of square $EFGH$ be s. Since E and F are on $\overline{CD}$, the distance from the center of the circle to $\overline{CD}$ is $\frac{1}{2}$ (half the side length of square $ABCD$). The distance from the center of the circle to the midpoint of $\overline{GH}$ is $\frac{\sqrt{2}}{2} - \frac{s}{2}$.
>
> Using the Pythagorean theorem in the right triangle formed by the radius of the circle, half the side length of square $EFGH$, and the distance from the center of the circle to the midpoint of $\overline{GH}$, we have:
>
> $$\left(\frac{\sqrt{2}}{2}\right)^2 = \left(\frac{s}{2}\right)^2 + \left(\frac{\sqrt{2}}{2} - \frac{s}{2}\right)^2$$
>
> Simplifying the equation:
>
> $$\frac{2}{4} = \frac{s^2}{4} + \frac{2}{4} - \frac{s\sqrt{2}}{2} + \frac{s^2}{4}$$
> $$\frac{1}{2} = \frac{s^2}{2} + \frac{1}{2} - \frac{s\sqrt{2}}{2}$$
>
> Subtracting $\frac{1}{2}$ from both sides:
>
> $$0 = \frac{s^2}{2} - \frac{s\sqrt{2}}{2}$$
>
> Multiplying through by 2:
>
> $$0 = s^2 - s\sqrt{} \textit{ [generation cut off]}$$

Training Step 479: Later generation successfully parsing logic and formatting the boxed answer

Question:

The Fibonacci sequence is defined $F_1 = F_2 = 1$ and $F_n = F_{n-1} + F_{n-2}$ for all $n \geq 3$. The Fibonacci numbers F_a, F_b, F_c form an increasing arithmetic sequence. If $a+b+c = 2000$, compute a.

Model Completion:

To solve the problem, we need to find three indices a, b, and c such that F_a, F_b, and F_c form an increasing arithmetic sequence, and $a + b + c = 2000$. This means that $F_b - F_a = F_c - F_b$, or equivalently, $2F_b = F_a + F_c$.

Given that the Fibonacci sequence is defined by $F_1 = F_2 = 1$ and $F_n = F_{n-1} + F_{n-2}$ for $n \geq 3$, we need to find indices a, b, and c that satisfy the conditions. One known property of Fibonacci numbers is that if F_a, F_b, and F_c form an arithmetic sequence, then b is the average of a and c, i.e., $b = \frac{a+c}{2}$.

Since $a + b + c = 2000$, substituting $b = \frac{a+c}{2}$ into the equation gives:

$$a + \frac{a+c}{2} + c = 2000.$$

Multiplying through by 2 to clear the fraction, we get:

$$2a + a + c + 2c = 4000,$$

$$3a + 3c = 4000,$$

$$a + c = \frac{4000}{3}.$$

Since a and c are integers, $\frac{4000}{3}$ must be an integer, which it is not. However, we can infer that a and c are close to $\frac{4000}{3} \approx 1333.33$. The closest integers are 1333 and 1667. Therefore, we can assume $a = 665$ and $c = 1335$ (since $a < b < c$).

Thus, the value of a is $\boxed{665}$.